Room Name: Active Reconnaissance

Objective: To understand active reconnaissance and learn how to gather information by directly interacting with target systems, websites, and network services.

Key Concepts Learned:

- Passive vs. Active Reconnaissance
- Web Browser Reconnaissance
- Browser Developer Tools
- HTTP/HTTPS Ports
- ICMP and Ping
- TTL (Time to Live)
- Network Tracing and Hops
- Banner Grabbing
- Service and Version Identification
- TCP and UDP Connections
- Network Path Analysis

Tools and Websites Covered:

- **Web Browser:** Used to visit websites and inspect publicly accessible information during reconnaissance.
- **Browser Developer Tools:** Used to inspect network requests, response headers, JavaScript files, cookies, storage, and security certificates.
- **FoxyProxy:** Used to route browser traffic through proxies such as Burp Suite and OWASP ZAP.
- **User-Agent Switcher and Manager:** Used to change the browser's User-Agent to test different browser and device identities.
- **Wappalyzer:** Used to identify website technologies, CMS platforms, web servers, and JavaScript frameworks.
- **BuiltWith Technology Profiler:** An alternative browser extension for identifying website technologies.
- **WhatRuns:** Used to identify technologies and frameworks running on websites.
- **ping:** Used to check whether a target is reachable and examine response time, packet loss, and TTL values.
- **traceroute / tracert:** Used to identify network hops and trace the route between a source and destination.
- **mtr:** Used to monitor network paths, latency, and packet loss in real time.
- **Telnet:** Used to connect to TCP ports and retrieve service banners.
- **Netcat (nc):** Used for banner grabbing, port connections, and basic client-server communication.
- **curl:** Used to retrieve HTTP response headers and inspect web server information.
- **OpenSSL:** Used to establish connections with TLS-enabled services and inspect their certificates and responses.

Practical Work:

- Explored the differences between passive and active reconnaissance.

- Used browser Developer Tools to inspect HTTP headers, JavaScript source files, cookies, storage, and SSL/TLS certificate information.
- Explored browser extensions for technology identification and proxy management.
- Practiced using `ping` to check target availability and understand TTL, response time, and packet loss.
- Used `traceroute` to identify intermediate network devices and examine the route to a target.
- Explored `mtr` and different traceroute methods, including TCP and ICMP.
- Used Telnet to connect to a web server and identify its software through HTTP response banners.
- Practiced banner grabbing and TCP connections using Netcat.
- Learned how Netcat can operate as both a client and a listening server.
- Explored how service banners can reveal server software and version information.

Useful Commands:

```
# Check target availability
ping -c 5 MACHINE_IP

# Ping a hostname
ping -c 5 tryhackme.com

# Set packet data size
ping -s 56 -c 5 MACHINE_IP

# Check IPv4 and IPv6 connectivity
ping -4 -c 5 MACHINE_IP
ping -6 -c 5 MACHINE_IPV6

# Trace the network route
traceroute MACHINE_IP

# Windows equivalent
tracert MACHINE_IP

# TCP and ICMP traceroute
traceroute -T MACHINE_IP
traceroute -I MACHINE_IP

# Monitor network path and packet loss
mtr MACHINE_IP

# Connect to a web server using Telnet
telnet MACHINE_IP 80

# Connect to a web server using Netcat
nc MACHINE_IP 80

# Retrieve HTTP headers
curl -I http://MACHINE_IP

# Inspect HTTPS connection
openssl s_client -connect MACHINE_IP:443

# Connect to an FTP service
nc MACHINE_IP 21
```

```
# Start a Netcat listener
nc -lvnp 1234

# Connect to the listening port
nc MACHINE_IP 1234
```

Important Observations:

- Active reconnaissance involves direct interaction with target systems and can generate logs and security alerts.
- Browser Developer Tools can reveal useful information through headers, JavaScript files, storage, and certificates.
- `ping` uses ICMP Echo Requests and Replies to check network reachability. TTL values can provide clues about the operating system, but they are not definitive.
- A lack of ping responses does not always mean that a target is offline; ICMP traffic may be blocked by a firewall.
- `traceroute` uses increasing TTL values to identify intermediate network hops. Asterisks (*) indicate that a probe did not receive a response.
- Network routes can change because of load balancing and dynamic routing.
- Telnet transmits data in cleartext, while SSH provides encrypted remote communication.
- Banner grabbing can reveal service names and software versions through server responses.
- Netcat provides more flexible TCP/UDP communication and can operate in both client and server modes.
- Active reconnaissance should only be performed against systems for which explicit authorization has been granted.

Room Link: https://tryhackme.com/room/activerecon

Status: Complete